# QUESTIONS BANK 1

**1.Homeostasis**

A must always be restored using negative feedback mechanisms
B provides for the tight regulation of all physiological variables✅
C is the sum of all chemical reactions in the body
D is a combination of positive and negative feedback mechanisms

**2.A negative feedback mechanism comprises**

A detectors, comparators, a fixed set point and effectors
B detectors, amplifiers, comparators, a set point and effectors
C detectors, attenuators, comparators, a set point and effectors
D detectors, comparators, a variable set point and effectors✅

**3.Positive feedback**

A does not exist in physiological systems
B is seen in the initiation of an action potential when sodium entry causes depolarization which in turn leads to potassium exit
C is unstable and requires some mechanism to break the feedback loop✅
D is switched off by negative feedback mechanisms

**4.The primary structure of a protein**

A is determined by the sequence of amino acids✅
B is a result of folding in the amino acid chain
C is overseen by molecular chaperones such as heat shock proteins
D has no effect on the functionality of the final molecule

**5.The percentages of water in each of the fluid compartments are approximately __ intracellular fluid, __ interstitial fluid and __ plasma**

A 22%, 13%, 65%
B 70%, 20%, 10%
C 65%, 22%, 13%✅
D 20%, 10%, 70%

**6.Osmosis is**

A the active movement of water across a permeable membrane from a region of high solute concentration to that of a lower solute concentration
B the passive movement of water across a semi-permeable membrane from a region of high solute concentration to that of a lower solute concentration
C the active movement of water across a permeable membrane from a region of low solute concentration to that of a higher solute concentration

D the passive movement of water across a semi-permeable membrane from a region of low solute concentration to that of a higher solute concentration✅

**7.The colloidal osmotic or oncotic pressure is**

A important for controlling cell volume
B a key determinant for fluid movement across capillary walls✅
C largely determined by the Na+ concentration in plasma
D higher in interstitial fluid than in plasma

**8.If a hypotonic fluid is ingested**

A intracellular fluid is diluted and cells swell✅
B intracellular fluid is diluted but cells stay the same size
C intracellular fluid is unaffected
D intracellular fluid is concentrated and cells shrink

**9.Most cell membranes are composed principally of**

A DNA and ATP
B proteins and lipids✅
C chitin and starch
D nucleotides and amino acids

**10.Which of the following correctly matches the organelle with its function?**

A mitochondrion — anaerobic respiration
B nucleus — protein assembly
C Golgi apparatus — glycosylation✅
D lysosome — movement

**11.Aerobic resynthesis of ATP occurs in the**

A mitochondria✅
B cytosol
C lipid membrane
D sarcoplasmic reticulum

**12.G-protein-coupled receptors**

A can activate signal proteins called G-proteins✅
B possess five membrane spanning segments
C can detect signal molecules, such as neurotransmitters or hormones, in the intracellular fluid
D penetrate the entire thickness of the phospholipids bilayer, with the intramembrane segments made up of chains of hydrophilic amino acid residues

**13.The sodium–potassium ATPase (sodium pump) is**

A a symporter
B an uniporter

C an antiporter✅
D an example of secondary active transport

**14.Which of the following is characterized by carrier-mediated transport down a chemical concentration gradient?**
A active transport
B facilitated diffusion✅
C diffusion
D osmosis

**15.Which of the following statements is NOT true about ligand-gated ion channels?**
A Neurotransmitters can act as external triggers to open ligand-gated ion channels
B Ligand-gated ion channels are found in the cell membrane
C Ligand-gated ion channels can be activated by cyclic AMP
D Differences in membrane potential can affect whether ligand-gated ion channel receptors open or close✅

**16.Ion channels**
A are always selective for a particular ion
B are opened and closed by a process called gating which is brought about by changes in the conformation of protein subunits✅
C cannot be gated by pH changes in the external environment
D provide a charged, hydrophobic pore through which ions can diffuse across the lipid bilayer

**17.If the extracellular potassium concentration rises from 4 mmol/L− to 5 mmol/ L, what happens to the resting membrane potential?**
A it becomes more negative
B it becomes less negative✅
C it remains the same
D it initiates an action potential

**18.In the resting membrane potential state**
A there are more Na+ than K+ channels in an open state
B the permeability to K+ ions exceeds that of Na+ ions✅
C the K+:Na+ permeability ratios are the same in both excitable and non-excitable cells
D the exact potential can be calculated using the Nernst equation

**19.During the rising phase of an action potential**
A voltage-gated Na+ channels open✅
B voltage-gated K+ channels open
C voltage-gated Na+ channels close
D voltage-gated K+ channels close

**20.For an action potential to occur**

A the generator potential must reach or exceed threshold✅

B the membrane potential must be out of the relative refractory period

C Na+ efflux must exceed K+ influx

D all the above

**21.Which is the correct sequence of events that follow a threshold potential?**

**1 the membrane becomes depolarized**

**2 sodium channels open and sodium ions diffuse inward**

**3 the membrane becomes repolarized**

**4 potassium channels open and potassium ions diffuse outward while sodium is actively transported out of the cell**

A 3, 2, 4, 1

B 2, 1, 4, 3✅

C 2, 1, 3, 4

D 1, 2, 4, 3

**22.Which of the following is true about the conduction of action potentials?**

A thicker axons conduct action potentials faster because there is a faster current flow in thicker axons

B action potentials travel faster in unmyelinated nerve fibres than myelinated nerve fibres

C the action potential in myelinated axons is generated only at the nodes of Ranvier✅

D the action potential in an unmyelinated nerve has an amplitude that declines with distance from its site of origin

**23.Saltatory conduction**

A can be up to 100 times faster than the conduction along the fastest unmyelinated axon

B occurs because the cell membrane beneath the myelin sheath has a higher density of Na+ channels

C is where depolarization jumps from one node of Ranvier to another and by doing so uses up more energy than in unmyelinated conduction

D can occur both orthodromically and antidromically if the axon is stimulated in its middle✅

**24.Nerve fibres that conduct at between 70 and 120 m/s− are**

A motor to muscle spindles

B afferents from primary muscle spindles and motor to skeletal muscles✅

C large unmyelinated fibres

D afferents from Golgi tendon organs

**25.The compound action potential**

A obeys the all-or-nothing law

B is usually a multi-peaked response with the slowest conducting fibres seen in the first peak

C is usually a multi-peaked response with the fastest conducting fibres seen in the first peak✅
D can be recorded by using intracellular recording electrodes

**26.Ligand binding to G-protein coupled receptors**
A activates phospholipase C-γ
B causes dissociation of Gα-GDP from the receptor followed by exchange of GTP for GDP
C stimulates Gα-GDP to exchange GDP for GTP, before dissociating from the receptor✅
D always causes elevation of intracellular [Ca2+]

**27.Receptor tyrosine kinases**
A are the most common form of receptor
B activate phospholipase A
C phosphorylate tyrosine residues on SH2 domain proteins
D dimerize on ligand binding✅

**28.Phospholipase C**
A liberates arachidonic acid from membrane phospholipids
B generates the second✅ messenger diacylglycerol (DAG)
C is located within the cytosol
D is activated by Gαs G-protein coupled receptors

**29.Protein kinase C**
A is a serine/threonine kinase✅
B is the primary effector for β-adrenergic receptors
C of the novel isoform type are activated by Ca2+ and diacylglycerol
D activates myosin light chain phosphatase

**30.The autonomic nervous system**
A controls smooth muscle, skeletal muscle and cardiac muscle
B controls smooth muscle, cardiac muscle and glandular activity✅
C mainly provides the afferent pathways for involuntary control of organs
D is subdivided into parasympathetic and sympathetic systems that mostly work antagonistically on blood vessels

**31.Sympathetic neurones**
A originate in the ventral horn of the spinal cord
B release adrenaline at their terminals
C can release acetylcholine at their terminals✅
D have peripheral ganglia found close to or in the target organ

**32.Parasympathetic neurones**
A originate in the brain stem and then run via the vagus to the paravertebral ganglia
B release noradrenaline at their terminals which in turn activates adrenergic receptors

C cause vasodilatation in the blood vessels of the external genitalia and salivary glands✅
D coordinate the so-called 'flight or fight' response

**33.In the synapse**
A action potentials are usually transmitted from one neurone to another by electronic coupling
B transmitter substances are stored in vesicles that are about 200 nm in diameter
C the presynaptic axon terminates in a bulbous swelling called a bouton which is separated from the postsynaptic neurone by a synaptic cleft✅
D the arrival of an action potential at the nerve ending causes an influx of Na+, which in turn causes the fusion of the vesicles containing the transmitter substance with the cell membrane and the subsequent release of the neurotransmitter

**34.Erythrocyte production**
A occurs in the liver and spleen in neonates✅
B is impaired at high altitude
C is stimulated by hyperoxia
D is stimulated by erythropoietin released from the liver and spleen

**35.White blood cells**
A do not contain a nucleus
B have a lifespan of 100–130 days
C include lymphocytes which destroy bacteria by a process called phagocytosis
D can release mediators such as histamine that are involved in inflammatory processes✅

**36.Reticulocytes**
A normally comprise 1-2% of red cells in the blood✅
B are nucleated
C have a lifespan of 120 days
D are formed from normoblasts after leaving the bone marrow

**37.Which one of the following is NOT associated with a plasma protein?**
A copper transport
B thalassemia✅
C the immune system
D capillary filtration

**38.Platelets**
A release ADP after activation✅
B are also called megakaryocytes
C can only be activated via their glycoprotein receptors
D have dense vesicles containing histamine

**39.During primary homeostasis**
A fibrinogen is converted to fibrin
B the initial vasoconstriction does not require platelet activation✅
C thromboxane A2 causes platelet adhesion
D activation of lipoxygenase is a vital step

**40.Activation of thrombin**
A is triggered by thrombomodulin
B always requires the presence of activated platelets
C is Ca2+ independent
D depends on Factor X✅

**41.Fibrin**
A requires Factor XIII to polymerize
B is destroyed by active protein C
C binds plasminogen✅
D activates von Willebrand factor

**42.Adaptive immunity is**
A acquired and results in immunological memory✅
B specific and mediated by natural killer cells
C mediated by cytokines
D non-specific

**43.Which of the following is NOT true regarding inflammation?**
A characterized by pain, heat, redness and swelling
B it increases vascular permeability
C the changes are initiated by cytokines released by neutrophils✅
D migrating neutrophils play an important role

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**45.Which form of immunoglobulin is most likely to be involved in secretions?**
A IgE
B IgG
C IgA✅
D IgM

**46.T helper cells**
A respond to antigens of any type
B only recognize antigens presented by major histocompatibility complex type 1
C phagocytose pathogens
D coordinate the immune response via cytokines✅

**47.Passive diffusion**
A involves a carrier medium
B requires the expenditure of energy
C refers to movement down a concentration gradient✅
D is described by Darcy's law

**48.rate of diffusion in a solution is described by**
A Fick's law✅
B Darcy's law
C Poiseuille's law
D Laplace's law

**49.The flow through the majority of the cardiovascular system at rest**
A is laminar✅
B is turbulent
C is described by Poiseuille's law which states that flow is dependent on the pressure difference across the ends of a tube and the resistance provided by the tube
D is not affected by changes in the viscosity of blood

**50.If a small alveolus was connected to a large alveolus in the lungs in the absence of surfactant**
A the large alveolus would empty into the small alveolus
B nothing would change
C the small alveolus would empty into the large alveolus✅
D both alveoli would empty

**51.The preoptic region of the hypothalamus**
A contains mostly cold sensitive thermoreceptors
B controls shivering
C is regulated by thermoreceptors in the posterior hypothalamus
D compares core temperature to the set point✅

**52.The sweat glands are activated by**
A noradrenaline
B bradykinin
C acetylcholine✅
D local heating

**53.Fever occurs when**

A core temperature is above the set point
B prostaglandin E2 increases the set point✅
C thermoregulation dysfunctions
D cyclooxygenase is inhibited by a pyrogen

**54.Non-shivering thermogenesis**

A is activated by β1-adrenergic receptors in brown adipose tissue
B requires activation of UCP1 by free fatty acids✅
C involves reduced electron flux through the mitochondrial electron transport chain
D only occurs in neonates

**55.Why does a climber's arterial PCO2 fall at altitudes above 2500 m?**

A the amount of CO2 in the air is less
B to compensate for the alkalosis
C metabolism is reduced due to the low PO2
D the low arterial PO2 is stimulating ventilation✅

**56.Which of the following does Not occur after acclimatization to altitude?**

A blood pH returns to normal
B cerebral spinal fluid pH returns to normal
C arterial PCO2 returns to normal✅
D plasma [HCO3-] is less than normal

**57.At what altitude would the arterial PO2 be closest to normal if breathing 100% O2?**

A sea level
B 6000 m
C 12 000 m✅
D 15 000 m

**58.Why is the risk of kidney stones increased during micro-gravity?**

A urine production is reduced
B redistribution of fluids away from lower part of body
C increased Ca2+ load due to bone demineralization✅
D reduced aldosterone production

**59.The two principal contractile proteins in skeletal muscles are**

A actin and troponin
B actin and tropomyosin
C actin and myosin✅
D myosin and tropomyosin

**60.The trigger to initiate the contractile process in skeletal muscle is**

A calcium binding to tropomyosin
B ATP binding to the myosin cross bridges
C calcium binding to troponin✅
D potassium binding to myosin

**61.After calcium ions are released from the sarcoplasmic reticulum they**

A initiate an action potential
B bind to actin
C cause sodium channels to open in the sarcolemmal membrane
D bind to troponin✅

**62.The A-bands of skeletal muscle do not change their width during muscle contraction because**

A the thick filaments are not involved in the sliding filament theory
B the A-bands are the thick filaments themselves which do not shorten✅
C the A-bands extend beyond the sarcomere
D the A-bands are the thin filaments themselves which do shorten

**63.At the neuromuscular junction**

A the muscle membrane possesses muscarinic receptors
B there is a one-to-one transmission of excitatory impulses from the motor neurone to the muscle fibres it innervates✅
C the motor nerve endings secrete noradrenaline
D the typical summed end plate potential (EPP) is usually 10 times the potential necessary to trigger an action potential

**64.During isotonic contraction**

A the A-band shortens
B the sarcomere and the I-band shorten✅
C the A-band and the I-band shorten
D the sarcomere and the A-band shorten

**65.An action potential in the motor end plate rapidly spreads to the central portions of the muscle cells by means of the**

A Z-lines
B sarcoplasmic reticulum
C pore in the plasma membrane
D transverse tubules✅

**66.During isometric contraction of skeletal muscle**

A the I-bands shorten and the A-bands stay the same length
B the thick and thin filaments slide past each other
C the sarcomere length does not change✅

D none of the above

**67.Motor unit' refers to**

A a single motor neurone and all the muscle fibres it innervates✅

B a single muscle fibre and all the motor neurones that innervate it

C all the motor neurones supplying a single muscle

D a pair of antagonist muscles

**68.Which of the following is NOT true regarding the comparison of type I (slow oxidative) and type IIB (fast glycolytic) skeletal muscle fibres?**

A type I fibres have a richer capillary supply

B type I fibres have smaller cell bodies

C type I fibres fatigue more readily✅

D type I fibres have a smaller diameter

**69.The sustained contraction by skeletal muscles in which individual twitches cannot be detected is called**

A summation

B recruitment

C tetanus✅

D twitching

**70. All of the following are involved in producing graded contractions of an entire muscle EXCEPT**

A small motor units (those with fewer muscle fibres, e.g. 5) are utilized for fine movements whilst large motor units (those with more muscle fibres, e.g. 600) are utilized for gross movements

B discharging many motor units at one time (spatial recruitment) allows increased force to be generated

C discharging individual motor units at a more rapid rate (temporal recruitment) allows increased force to be generated

D contraction of only a portion of the muscle fibres within a single motor unit during a finely graded movement✅

**71.Myocytes are**

A approximately 100 μm × 50 μm in dimensions

B multinucleated

C linked to one another at their ends by electronic junctions called intercalated discs✅

D low in mitochondria

**72.Regarding smooth muscle**

A it has actin and myosin filaments organized into myofibrils

B it is multinucleated

C like cardiac muscle, the cells are joined together by gap junctions called intercalated discs

D it is capable of prolonged contraction without fatigue and with little energy consumption✅

**73.Which of the following statements is NOT true?**

A Smooth muscle has no troponin
B Smooth muscle is innervated by neurones of the autonomic nervous system
C Multiunit smooth muscle is made up of individual fibres not connected by gap junctions
D Smooth muscle cells are not affected by local tissue factors and hormones✅

**74.The effect of nerve stimulation on smooth muscle is**

A excitatory only
B inhibitory only
C both excitatory or inhibitory✅
D modulatory only

**78.The cardiovascular system**

A contains approximately 7 L of blood in a 70 kg man
B is arranged mostly in series with each tissue receiving blood from the aorta
C is arranged mostly in parallel with each tissue receiving blood from the aorta✅
D maintains a constant pressure throughout the syste

**79.Smooth muscle is not to be found in the walls of**

A veins
B venules
C arterioles
D capillaries✅

**80.In the heart, the cardiac output is approximately /min *at rest, rising to above* /min during exercise. The stroke volume at rest is approximately __.**

A 20 L, 100 L, 1 L
B 5 L, 100 L, 70 mL
C 500 mL, 20 L, 7 mL
D 5 L, 20 L, 70 mL✅

**81.The mean arterial blood pressure is**

A 110 mmHg in a healthy person
B calculated by averaging the systolic blood pressure and the diastolic blood pressure
C estimated as the diastolic blood pressure plus one third of the systolic blood pressure
D estimated as the diastolic blood pressure plus one third of the pulse pressure✅

**82.Blood flows from the right atrium into the right ventricle via**

A the mitral valve
B the semilunar valves
C the tricuspid valve✅

D the AV node

**83.Conduction of impulses between the atria and the ventricles is channelled through the**

A AV node✅
B annulus fibrosus
C SA node
D bundle of His

**84.The atrioventricular node**

A conducts impulses rapidly allowing almost immediate activation of ventricular muscle
B delays impulses for about 200 ms, allowing time for atrial contraction to complete ventricular filling
C is connected to specialized wide, fast conducting myocytes in the bundle of His and Purkinje fibres✅
D is a band of fibrous connective tissue which separates the atria from the ventricles

**85.Which of the following statements concerning coronary circulation is true?**

A Left ventricular perfusion only occurs during diastole
B The heart receives a rich blood supply from the left and right coronary arteries arising from the coronary sinus
C The coronary veins run parallel to the right coronary arteries and empty into the aortic sinus
D During systole, contraction of the ventricles compresses the coronary arteries and suppresses blood flow✅

**86.The aortic valve**

A prevents the backflow of blood into the aorta during ventricular diastole
B prevents the backflow of blood into the left ventricle during ventricular diastole✅
C prevents the backflow of blood into the left ventricle during ventricular systole
D prevents the backflow of blood into the aorta during ventricular systole

**87.Which of the following indicates the causes of the first and second heart sound in the correct order?**

A atrial systole — ventricular systole
B semilunar valve closure — atrioventricular valve closure
C ventricular diastole — semilunar valve closure
D ventricular systole — ventricular diastole✅

**88.During which phases of the cardiac cycle do the atrioventricular valves remain open?**

A atrial diastole
B isovolumetric ventricular relaxation
C isovolumetric ventricular contraction
D passive filling✅

**89. The ventricular end-diastolic volume in man is approximately __ and the end-diastolic pressure is __**

A 130 mL, >100 mmHg
B 130 mL, <10 mmHg✅
C 70 mL, <10 mmHg
D 70 mL, >100 mmHg

**90.Action potentials in ventricular muscles**

A are identical to those in skeletal muscles except for the duration of the action potential
B have a plateau phase caused by the delay in the opening of K+ channels
C are initiated when ventricular myocytes are depolarized to a threshold potential of −50 mV
D have a refractory period which prevents another action potential being initiated until the muscle relaxes✅

**91.The cells of the sinoatrial node**

A have a resting potential of −90 mV
B have action potentials which exhibit a slow upstroke because of the presence of L-type calcium channels✅
C are the only cells in the heart that can act as pacemaker cells
D are directly affected by noradrenaline and acetylcholine in that they slow down and speed up the heart respectively

**92.Excitation–contraction coupling in cardiac ventricular cells requires**

A efflux of Na+ ions
B efflux of K+ ions
C influx of Ca2+ ions✅
D influx of Cl− ions

**93.Noradrenaline**

A has a positive inotropic effect on the heart muscle cells whilst acetylcholine has a negative inotropic effect
B is released by the sympathetic fibres innervating the SA node only
C has a positive chronotropic effect on cardiac cells by increasing the rate of decay of the pacemaker potential✅
D slows down the Ca2+ sequestration into the sarcoplasmic reticulum thereby increasing contractility of the cardiac cells

**94.Cardiac output is**

A the volume of blood pumped per minute by both ventricles
B the volume of blood flowing through the systemic circulation per minute✅
C not affected by the filling pressure (preload)
D approximately 5 L/min at rest rising to over 100 L/min during exercise

**95.Starling's law of the heart**

A states that 'the stroke volumes of the left and right ventricles are matched'

B concerns the relationship between the degree of stretch of cardiac muscle and the force of contraction✅

C causes an increase of contractility of cardiac muscle

D can equally be applied to skeletal as well as cardiac muscle

**96.Constriction of the veins**

A decreases venous compliance and therefore increases CVP

B increases venous resistance and therefore decreases CVP

C increases the slope of the vascular function curve✅

D reduces venous return

**97.Which of the following statements is NOT true?**

A The mechanism underlying Starling's law of the heart is intrinsic to cardiac muscle

B The autonomic nervous system provides an important extrinsic influence on cardiac output

C Sympathetic stimulation can cause an increase in contractility, which can be an increase in force without a change in length

D Cardiac output is only affected by the filling pressures of the left side of the heart and the effects of the autonomic nervous system on the heart rate and contractility✅

**98.Smooth muscle and elastin filaments are found in the**

A tunica intima

B tunica adventitia

C tunica media✅

D elastic lamina

**99.Fenestrated capillaries**

A are more permeable than continuous capillaries and are found in the skin and muscle

B are less permeable than continuous capillaries and are found in endocrine glands and intestinal villa

C have less tight junctions than continuous capillaries and have pores in their endothelial cells✅

D are found in bone marrow, liver and spleen

**100.The following substances are all vasodilators**

A angiotensin II, histamine and bradykinin

B histamine, bradykinin and substance P✅

C acetylcholine, bradykinin and endothelin-1

D substance P, noradrenaline and phospholipase C

**101.The endothelium**

A can synthesize a number of important vasoconstrictors such as nitric oxide (NO)

B can synthesize a number of important vasodilators such as endothelium-derived relaxing factor (EDRF)✅

C can only synthesize vasodilators

D plays a minor role in the regulation of vascular tone

**102.An increase in the mean arterial blood pressure may result from**

A an increase in peripheral resistance

B an increase in the heart rate

C an increase in venous return

D a, b and c✅

**103.Baroreceptors**

A in the carotid and aortic bodies send impulses via the glossopharangeal and vagus nerves to the medulla of the brain stem

B are sensors for mean arterial blood pressure that respond to stretching of the carotid sinus and aortic arch✅

C do not exhibit adaptation

D are most sensitive between 120 and 180 mmHg

**104.A fall in mean arterial blood pressure**

A causes an increase in baroreceptor activity and a reduction in renal perfusion pressure

B along with sympathetic stimulation will activate the renin–angiotensin system and the production of angiotensin II✅

C increases Na+ and water excretion

D is sensed by atrial baroreceptors

**105.Which of the following statements is NOT true?**

A 20% of the blood volume can be lost without significant problems

B An acute fall in mean arterial blood pressure can result from a sudden blood loss, a profound vasodilatation or an acute failure of the heart to pump blood

C Following the loss of less than 20% of the blood volume, the volume of blood is restored within minutes of the blood loss due to arteriolar constriction✅

D A loss of 30–50% of blood volume can be survived if a blood transfusion is given within an hour

**106.Which of the following substances are highly lipophilic and therefore can cross the endothelial lipid bilayer membrane easily?**

A oxygen and carbon dioxide✅

B water

C glucose and Na+ ions

D plasma proteins

**107.In a capillary, the net flow of water between the blood and the interstitial fluid can change in its direction between the arterial and venous ends of the capillary. This difference at the two ends reflects a change in which of the following factors?**

A osmotic pressure
B vessel diameter
C blood velocity
D blood pressure✅

**108.Lymph fluid is returned into the general circulation via the**

A left atrium
B superior vena cava
C subclavian vein✅
D azygous vein

**109.Which of the following statement is NOT true?**

A Oedema is the swelling of the tissues due to excess fluid in the interstitial space
B Oedema can be caused by an increased venous pressure
C The colloidal osmotic pressure normally varies between
~25 mmHg at the arteriolar end of the capillary and ~15 mmHg at the venous end✅
D Lymphatic capillaries are blind-ended bulbous tubes approximately 15–75 μm in diameter

**110.Autoregulation is the ability of a tissue to maintain a constant blood flow in the face of variations of pressure. It is particularly important in the**

A brain, kidneys and heart✅
B skin and gut
C skeletal muscle
D lungs

**111.The following substances are autocoids (i.e. local hormones)**

A bradykinin, nitric oxide and histamine
B serotonin, histamine and adenosine
C bradykinin, histamine and serotonin✅
D nitric oxide, adenosine and carbon dioxide

**112.Which of the following statements is true?**

A Skeletal muscle comprises approximately 20% of body weight yet in exercise can take over 80% of cardiac output
B In skeletal muscles at rest, the majority of capillaries are not perfused, as their arterioles are constricted✅
C In skeletal muscle, capillaries are recruited during exercise by metabolic hyperaemia caused by the release of $CO_2$ and $Ca^{2+}$
D The endothelial cells of the capillaries in the brain have very few tight junctions so there is free movement of fluids between the blood and the cerebrospinal fluid

**113.Blood flow**

A in the brain is controlled by both sympathetic and parasympathetic nerve fibres

B through skeletal muscle involves arteriovenous anastomoses which directly link arterioles and venules, by-passing the capillaries

C in the pulmonary circulation is not controlled by autonomic nerves or metabolic products✅

D in the pulmonary circulation is increased during hypoxia due to vasodilatation in small arteries

**114.The U-shaped rings that form the framework of the trachea and help to keep it open are composed of**

A skeletal muscle

B bone

C cartilage✅

D fibro-elastic tissue

**115.Alveolar type II pneumocytes are**

A cuboidal cells that secrete surfactant✅

B squamous cells involved in gas exchange

C ciliated cells that move the mucus

D columnar cells that secrete mucus

**116.Which muscles are involved in quiet inspiration?**

A the internal and external intercostal muscles

B the diaphragm

C the abdominal muscles

D the diaphragm and external intercostal muscles✅

**117.In order for the lungs to function normally the intrapleural pressure must**

A alternate between being less than or greater than atmospheric pressure

B be the same as atmospheric pressure

C be between 0.2 and 0.5 kPa

D be between −0.2 and −0.5 kPa✅

**118.Which of the following statements is true?**

A The change in the volume of the lungs with pressure is a measure of their compliance✅

B The total compliance of the chest is determined solely by the compliance of the lungs

C The recoil of the lungs assists inspiration

D Pulmonary surfactant maintains a constant low surface tension in the alveoli

**119.Which of the following statements is/are true?**

**1 Neonatal respiratory distress syndrome is caused by lack of sufficient surfactant**

**2 Surfactant is produced by type I pneumocytes**

**3 Surfactant creates hysteresis**
**4 The pressure due to surface tension is greater in a soap bubble of greater radius**
A 1 only✅
B 3 only
C 2 and 4
D 1, 3 and 4

**120.Dynamic compression of airways**
A is caused by bronchoconstrictors such as histamine, prostaglandins and leukotrienes
B occurs during forced expiration✅
C is brought about by reflex activation of parasympathetic nerves
D does not occur in normal expiration because the intrapleural pressure remains positive throughout

**121.Peak expiratory flow rate (PEFR)**
A is not dependent on initial lung volume
B provides as much information as the forced expiratory volume against time
C increases as the airways resistance increases
D is useful when following the progression and treatment of a disease such as asthma✅

**122.According to Dalton's law the partial pressure of O2 (Po2) in dry air at a barometric pressure of 101 kPa (sea level) is __ kPa and at a barometric pressure of 34 kPa (summit of Everest) is __ kPa**
A 21.2, 7.14✅
B 21.0, 8.0
C 19.9, 5.8
D 20, 6.15

**123.Typical partial pressures for alveolar air at 37°C and 100% humidity are Pco2 __ kPa, Po2 __ kPa, Pn2 __ kPa and saturated water vapour pressure __ kPa**
A 0, 19.9, 74.8, 6.3
B 0.05, 0.13, 0.75, 0.06
C 5.3, 13.3, 76.1, 6.3✅
D 0, 21.2, 79.8, 0

**124.The quantity of gas dissolving in a fluid is described by**
A Dalton's law
B Boyle's law
C Charles' law
D Henry's law✅

**125.CO2 crosses the alveolar–capillary membrane**
A faster than O2 because it has a greater molecular weight
B slower than O2 because it has a greater molecular weight

C faster than O2 because it is more soluble in biological membranes✅
D slower than O2 because it is less soluble in biological membranes

**126.The O2–haemoglobin dissociation curve is shifted to the right by**
A a decrease in acidity
B an increase in Pco2✅
C a decrease in temperature
D a decrease in 2,3-DPG

**127.CO2 is transported in the blood approximately**
A 60% as bicarbonate, 30% as carbamino compounds and 10% dissolved✅
B 60% as bicarbonate, 20% as carbamino compounds and 20% dissolved
C 50% as bicarbonate, 40% as carbamino compounds and 10% dissolved
D 50% as bicarbonate, 35% as carbamino compounds and 15% dissolved

**128.Which of the following statements is NOT true?**
A Fetal haemoglobin binds 2,3-DPG less strongly than does adult haemoglobin, thus shifting the dissociation curve to the right✅
B Fetal haemoglobin facilitates the transfer of oxygen from maternal blood to the fetus
C Carbon monoxide binds 240 times more strongly than oxygen to haemoglobin
D In anaemia, arterial Po2 and O2 saturation remain normal

**129.Hyperventilation**
A is rapid breathing during exercise
B reduces the arterial Pco2 and increases arterial Po2 substantially
C can lead to light-headedness, visual disturbances and tetanus
D can be caused by pain, hysteria and strong emotion✅

**130.In the control of breathing**
A afferents from the stretch receptors feed into the pneumotaxic centre in the pons to modulate breathing
B when expiratory neurones fire they inhibit the activity of inspiratory neurons✅
C voluntary control of breathing is possible because motor neurones from the cerebral cortex directly innervate respiratory group neurones in the medulla, altering their activity
D juxtapulmonary receptors detect fluid in the alveolar walls to cause an increased rate of breathing and an increase of both heart rate and blood pressure

**131.Ventilation may be increased**
A by a decrease in plasma pH (acidosis)✅
B by breathing O2-enriched air
C by stimulation of the intracranial chemoreceptors by hypoxia
D by stimulation of the carotid sinus by hypercapnia

**132.The receptors that are located on the alveolar and bronchial walls close to the capillaries are called**

A pulmonary stretch receptors
B irritant receptors
C proprioceptors
D juxtapulmonary receptors✅

**133.The central chemoreceptor**

A is a collection of neurones near the ventrolateral surface of the pons
B responds to changes in both blood Pco2 and Po2
C responds to a rise in H+ in the CSF and leads to an increase in ventilation✅
D is responsible for only 20% of the response to CO2 in humans

**134.In a situation where there is a ventilation–perfusion mismatch with a VA/Q value of less than unity, i.e. shunt effect**

A there is no gas exchange
B the blood has a lower than normal Po2, and a higher than normal Pco2✅
C the blood has a higher than normal Po2, and a lower than normal Pco2
D there is excessive ventilation

**135.Ventilation with O2-enriched air will**

A improve oxygenation in regions of high VA/Q
B improve oxygenation in regions of low VA/Q✅
C improve oxygenation in shunts
D have no effect of VA/Q in any situation

**136.Gravity**

A causes a higher blood flow at the base of the lungs than at the apex of the lungs✅
B causes a less negative intrapleural pressure at the apex of the lungs than at the base of the lungs
C causes ventilation to be greater at the apex of the lungs than at the base of the lungs
D has no effect on VA/Q

**137.Anatomical right to left shunts where oxygenated blood from the lungs is diluted by venous blood**

A occur only in disease or congenital heart malformations
B account for approximately 5% of cardiac output in healthy individuals
C commonly result in a low arterial Po2 and a high Pco2
D can occur in diseases in which regions of the lungs are not ventilated✅

**138.Which one of these statements is false? The nephron**

A has endocrine functions
B is under endocrine control
C absorbs more ions and molecules than it secretes

D lies exclusively within the renal cortex✅

**139.In the kidney**

A 85% of the nephrons are juxtamedullary nephrons and 15% are cortical nephrons

B the proximal, distal and collecting ducts are found in the medulla

C the proximal tubule walls are formed of thin, flat squamous epithelial cells with no microvilli

D urine is propelled through the ureter into the bladder by peristalsis✅

**140.Which of these statements is true?**

A The kidney receives approximately 30% of cardiac output

B The capillaries of the glomerulus are the site of filtration and drain into the efferent vein

C The only blood supply to the medulla is the vasa recta✅

D The blood flow through the cortex and the medulla are approximately equal

**141.Micturition is**

A totally controlled by higher voluntary centres

B initiated by a spinal reflex when urine pressure reaches a critical level✅

C brought about by the contraction of skeletal muscle (the detrusor muscle) surrounding the bladder

D totally controlled by the internal and external urethral sphincters

**142.In the glomerulus, if the diameter of the afferent arteriole is smaller than the diameter of the efferent arteriole, then**

A the net filtration pressure will decrease✅

B blood pressure in the glomerulus will increase

C GFR will increase

D there will be little change in GFR

**143.Glomerular filtration rate is approximately __ mm/min, whereas renal plasma flow is approximately __ mL/min**

A 50 300

B 125 600✅

C 200 300

D 125 250

**144.Which of the following does NOT reduce glomerular filtration rate?**

A renal disease

B local vasoconstrictors

C sympathetic activation

D angiotensin II✅

**145.Glomerular filtration rate can be calculated using any substance that**
A is freely filtered by the nephron
B is freely filtered and secreted by the nephron
C is freely filtered and neither reabsorbed nor secreted by the nephron✅
D is freely filtered and completely reabsorbed by the nephron

**146.Reabsorption and secretion in the proximal tubule involves**
A diffusion through tight junctions and lateral intercellular spaces
B active transport involving proteins called transporters
C facilitated diffusion
D all of the above✅

**147.Which of the following is not reabsorbed in the proximal tubule?**
A glucose
B amino acids
C organic acids✅
D bicarbonate

**148.The tubular transport maximum (Tm) for glucose is approximately**
A 21 mg/min
B 21 mmol/min✅
C 11 mmol/min
D 380 mmol/min

**149.In the proximal tubule, water**
A is actively reabsorbed
B is actively secreted
C reabsorption increases tubular concentrations of $Cl^-$, $K^+$, $Ca^{2+}$ and urea✅
D content is decreased by about 50%

**150.In which part of the nephron does antidiuretic hormone (ADH) have its main effect?**
A the proximal tubule
B the descending limb of the loop of Henle
C the ascending limb of the loop of Henle
D the distal tubule and the collecting duct✅

**151.Fluid entering the loop of Henle is normally __ with plasma, and that entering the distal tubule is normally __ with plasma.**
A isotonic, hypertonic
B hypertonic, hypotonic
C isotonic, hypotonic✅
D hypotonic, hypertonic

**152.Which of the following is NOT true of the countercurrent multiplier system?**

A The ascending limb of the loop of Henle transports Na+ and Cl− by active transport

B The descending limb of the loop of Henle is impermeable to water✅

C The fluid of the ascending limb of the loop of Henle becomes relatively dilute (low osmolarity)

D The vasa recta removes water reabsorbed from both the loop of Henle and the collecting ducts

**153.At its most concentrated, final urine can be about __ times more concentrated than plasma**

A 2

B 8

C 5✅

D 10

**154.Plasma osmolality is**

A decreased in water deficiency

B increased by the ingestion of water

C controlled by osmoreceptors in the anterior pituitary

D regulated by antidiuretic hormone (ADH)✅

**155.Antidiuretic hormone (ADH)**

A is a nine-amino-acid peptide formed from a precursor synthesized in the pituitary

B is stored in secretory granules in the anterior pituitary

C is released from secretory granules by action potentials from osmoreceptors✅

D causes vasoconstriction via V2 receptors

**Which of the following statements is NOT true?**

156.A The control of body Na+?content by the kidney is the main regulator of body fluid volume

B Stretch receptors on the low pressure side of the heart detect changes in the central venous pressure which reflect changes in blood volume

C Increased pressure in the renal afferent arterioles stimulates renin release✅

D Angiotensin II is the primary hormone for Na+?homeostasis

**157.Atrial natriuretic peptide (ANP)**

A is released from left atrial muscle in response to stretching caused by increased arterial blood pressure

B suppresses the production of renin and aldesterone but not ADH

C inhibits the effects of ADH in the distal nephron and causes renal vasodilatation✅

D causes increased excretion of Na+ but not water

**158.The Henderson–Hasselbalch equation**

A allows one to determine the molecular weight of a weak acid from its pH alone

B is equally useful with solutions of acetic and hydrochloric acid

C employs the same value for pK for all weak acids

D relates the pH of a solution to the pK and the concentrations of weak acid and conjugate base✅

**159.Which of the following statements about buffers is true?**

A The pH of a buffered solution remains constant no matter how much acid or base is added to the solution

B The strongest buffers are those composed of strong acids and strong bases

C The maximal buffering capacity occurs when pH = pK✅

D The pH of blood is not maintained by a buffering system

**160.In the proximal renal tubule**

A bicarbonate is freely filtered✅

B over 95% of bicarbonate is reabsorbed

C bicarbonate is transported into the interstitium largely by $Na^+$–$HCO_3^-$ antiporters

D there is a net absorption of H?

**161.Which of the following statements is NOT true?**

A In the proximal nephron, $H^+$ secretion promotes $HCO_3^-$ reabsorption

B A fall in blood pH will stimulate renal $H^+$ secretion

C Renal mechanisms are fast because their capacity to handle $H^+$ and $HCO_3^-$ is greater than that of the lungs for handling $CO_2$✅

D Hypokalaemia is associated with an enhanced $H^+$ secretion

**162.The gastrointestinal tract**

A is made up of the mouth, oesophagus, stomach, pancreas, liver, and small and large intestine

B general structure comprises the mucosal layer, the lamina propria, the muscularis mucosa, the submucosal layer, followed by a layer of longitudinal muscle and then a layer of circular muscle

C has two layers of smooth muscle between which lies a nerve plexus called Meissner's plexus

D has two nerve plexuses called the submucosal plexus (Meissner's plexus) and the myenteric plexus (Auerbach's plexus)✅

**163.Mastication**

A is necessary for most foods to be fully absorbed by the rest of the gastrointestinal tract

B usually develops forces of between 1500 and 2000 N between the teeth

C can lead to stimulation of periodontal and mucosal mechanoreceptors that in turn lead to the reflex secretion of saliva from the submandibular/sublingual salivary glands but not from the parotid glands
D involves the coordinated activity of the teeth, jaw muscles, temporomandibular joint and tongue as well as the lips, palate and salivary glands✅

**164.Which of the following statements is NOT true? Saliva**
A normally has a very low pH to enhance breakdown of food in the mouth✅
B aids swallowing
C contains both inorganic and organic constituents
D is hypotonic to plasma when secreted at a low rate

**165.Swallowing**
A involves both voluntary and reflex responses, initiated by the stimulation of mechanoreceptors with afferents in the glossopharyngeal (IX) and trigeminal (V) nerves
B is coordinated by a group of neurones called the swallowing centre found in the thalamus
C occurs after a normal expiration before inspiration begins again
D involves a complex series of events in which the soft palate elevates to prevent food entering the nasal cavity, the larynx is raised and the glottis is closed and the food pushes the tip of the epiglottis over the tracheal opening to prevent food entering the trachea✅

**166.Gastric secretion is**
A increased by anticipation of a meal✅
B increased by acid in the duodenum
C decreased by gastrin
D increased by secretin

**167.With regard to gastric secretion**
A the cephalic phase is mainly brought about by the thought of food
B the gastric phase is mainly stimulated by activity of the vagus nerve
C mechanoreceptors in the stomach wall can be stretched and set up local myenteric reflexes and longer vagal reflexes which stimulate the release of gastrin, histamine, acid, enzymes and mucus✅
D a high pH in the stomach inhibits gastrin secretion

**168.Which of the following statements is NOT true?**
A The gastric phase of secretion normally lasts for about 3 hours
B The chyme enters the duodenum through the pyloric sphincter
C The presence of chyme in the pyloric antrum causes opening of the pyloric sphincter
D The rate at which the stomach empties depends on the volume in the antrum and the fall in pH of the chyme, both leading to a decrease in emptying✅

**169.Which of the following is NOT a function of the stomach?**
A to store food temporarily

B to digest food using acids, enzymes and peristaltic movements
C to regulate the release of chyme into the small intestine
D to secrete intrinsic factor which is essential for the absorption of fat-soluble vitamins✅

**170.The parts of the small intestine (start with the part closest to the stomach) are**
A duodenum, jejunum, ileum✅
B duodenum, ileum, jejunum
C ileum, jejunum, duodenum
D ileum, duodenum, jejunum

**171.The intestinal phase of gastric juice secretion**
A is brought about by both distension of the duodenum and the release of secretin
B is probably due to activation of✅ G-cells in the intestinal mucosa
C is activated by the release of secretin and cholecystokinin
D controls gastric emptying

**172.Which of the following is NOT secreted by the intestinal epithelial cells (brush border)?**
A trypsin✅
B maltases
C sucrase
D lactase

**173.All of the following are required for fat digestion except**
A bile acids
B pancreatic lipase
C cholic acid
D monoglycerides✅

**174.With regard to pancreatic secretion**
A Pancreatic acinar cells contain trypsin
B When acid enters the duodenum it stimulates pancreatic secretion✅
C Pancreatic secretion is inhibited by gastrin secreted by the G-cells of the antrum
D Cholecystokinin inhibits secretion from the exocrine pancreas

**175.Which of the following statements is true?**
A Bile is diluted in the gallbladder
B Bile salts are mostly absorbed in the distal ileum✅
C Bile salts are formed from the breakdown products of haemoglobin
D The gallbladder can produce 500–1000 mL of bile per day

**176.The liver**
A consists of four lobules

B is involved in metabolism of a vast number of substances produced by the digestion and absorption of food as well as having an important endocrine function
C contains a single column of cells called hepatocytes which secrete hepatic bile which is hypotonic with plasma
D hepatocytes secrete hepatic bile which contains bile salts, bile pigments, cholesterol, lecithin and mucus✅

**177.Cholecystokinin**
A is a glycoprotein
B is secreted mainly by the distal ileum
C stimulates relaxation of the gallbladder
D is released into the bloodstream and stimulates the secretion of pancreatic enzymes✅

**178.The parts of the large intestine (starting from the part closest to the small intestine) are the**
A ascending, transverse, descending and sigmoid colon, caecum, rectum and anal canal
B caecum, descending, transverse, ascending and sigmoid colon, rectum and anal canal
C caecum, ascending, transverse, descending and sigmoid colon, anal canal and rectum
D caecum, ascending, transverse, descending and sigmoid colon, rectum and anal canal✅

**179.In the large intestine**
A the caecum, and ascending and descending colon are innervated by parasympathetic branches of the vagus nerve
B the descending and sigmoid colon, rectum and anal canal are innervated by the parasympathetic branches of the vagus nerve
C the external anal sphincter is composed of striated muscle and innervated by the pudendal nerve✅
D stimulation of the sympathetic nerve fibres causes segmental contraction

**180.The main function of the large intestine is**
A to store the residues of food and absorb water and electrolytes✅
B mixing and propulsion of chyme at a rate of 50–100 cm/h
C the further digestion and absorption of nutrients
D defecation

**181.The bacteria in the gastrointestinal tract are**
A evenly distributed throughout the small and large intestines
B found mainly in the large intestine because of the acid environment
C 99% anaerobic and most are lost in the faeces✅
D involved in the synthesis of vitamins B12 and C

**182.Which of the following is NOT a hormone?**
A G-proteins
B insulin

C cortisol✅
D adrenaline

**183.With regard to endocrine control**
A The endocrine system provides a rapid, precise but short term control of cellular function
B Paracrine cells release their hormones into the bloodstream so that they reach all parts of the body
C Autocrine cells release hormones that control their own function✅
D Hormones are secreted from either classical glandular tissues or whole organs and not from other tissues

**184.Hormones**
A are activated by metabolic transformation by enzymes in the liver or at the site of action
B rely heavily on a positive feedback mechanism for their control
C are involved in homeostasis, reproduction, growth and development and metabolism✅
D can exert their effect directly on the cell by interaction with non-specific receptor proteins

**185.Antidiuretic hormone (ADH) is secreted from the __ and its main target tissue is the __**
A anterior pituitary, kidney
B intermediate pituitary, hypothalamus
C posterior pituitary, kidney✅
D adrenal cortex, liver and kidney

**186.In an anabolic state**
A storage molecules are broken down
B there is an increase in glucagon and a decrease of insulin✅
C there is an uptake of glucose and fatty acids by the liver, muscle and adipose tissue
D leptin is released from adipose tissue

**187.The pancreatic cells that secrete somatostatin are the**
A F-cells
B A (or α) cells
C B (or β) cells
D D (or δ) cells✅

**188.The effects of insulin on target cells**
A are mediated by voltage gated Ca2+ channels
B are to stimulate glycogen uptake
C are to stimulate the release of glucagons
D are mediated by a receptor tyrosine kinase✅

**189.The effects of hypoglycaemia do NOT include**

A weakness
B blurred vision
C headache
D polyuria✅

**190.Which hypophyseal structure receives signals from the hypothalamus via the hypophyseal portal vessels?**
A neurohypophysis
B supraoptic nucleus
C adenohypophysis✅
D paraventricular nucleus

**191.Which of the following hormones is NOT produced by the anterior pituitary**?
A follicle-stimulating hormone
B thyroid-stimulating hormone
C oxytocin✅
D adrenocorticotrophic hormone

**192.Hypophysiotropic hormones**
A originate from the magnocellular neurones in the hypothalamus
B are peptides or proteins released by nerve endings into the bloodstream at the inferior hypophyseal artery
C are not involved in neuroendocrine reflexes✅
D originate from parvocellular neurones whose cell bodies are found in the anterior pituitary

**193.Which of the following statements is NOT true?**
A Hormones released from the hypothalamus tend to appear in the blood in discrete pulses and not as a continuous secretion
B The hypothalamus surrounds the second ventricle at the base of the medial forebrain✅
C The hypothalamic subdivisions important for endocrine function include the paraventricular, periventricular, supraoptic and arcuate nuclei.
D The release of ADH is controlled by conventional negative feedback mechanisms based on osmolality whereas oxytocin involves positive feedback mechanisms

**194.Concerning thyroid hormones**
A T3 is some 10 times more potent in the tissues than T4✅
B T4 and T3 are mainly transported in the blood bound to albumin
C T3 is converted into T4 in the tissues
D T4 and T3 are secreted by the thyroid in equal quantities

**195.The primary effect of T3 and T4 is to**
A decrease blood glucose
B promote the release of calcitonin
C promote heat generating metabolic reactions✅

D stimulate the uptake of iodine by the thyroid

**196.What hormone would best fit this description? 'affects metabolism of cells; necessary for the CNS to develop properly; necessary for normal bone growth; stored extracellularly'**

A cortisol
B growth hormone
C T4/T3✅
D thyrocalcitonin

**197.In mild to moderate cases of hypothyroidism the following symptom is not usually seen**

A lethargy
B sluggishness
C intolerance to cold
D tissue bloating✅

**198.Growth factors**

A stimulate mitosis, growth and apoptosis
B are modified amino acids
C are peptide hormones✅
D are not involved in the stimulation of red blood cell production

**199.With regard to insulin and insulin-like growth factors (IGF-1 and IGF-2)**

A They both have a similar function
B The IGFs are mitogenic, trophic and act as survival factors✅
C Insulin promotes catabolic activity generally
D They have quite different structures

**200.With the exception of __, growth factors work by activating receptor tyrosine kinases**

A TGFα, TGFβ and EGF
B TGFβ, erythropoietin and EGF
C TGFα, erythropoietin and the cytokines✅
D TGFβ, erythropoietin and the cytokines

**201.The following growth factor has been implicated in the maintenance of colorectal and breast cancers**

A EGF✅
B FGF-1–24
C PDGF
D NGF

**202.With regard to growth hormone (somatotrophin), which of the following statements is NOT true?**

A Growth hormone provides the main drive for growth spurts during development

B Growth hormone is a protein released from the anterior pituitary somatotrophs under the control of the hypothalamus

C The release of growth hormone varies throughout the day, with the peak following a large meal✅

D Growth hormone release is increased in response to exercise and stress

**203.The overproduction of growth hormone in adults leads to**

A gigantism

B dwarfism

C acromegaly✅

D osteoporosis

**204.The type of bone that has a dense structure and provides most of the strength of the skeleton is called**

A trabecular bone

B cortical bone✅

C diaphyseal bone

D epiphyseal bone

**205.The large cells present in mature bone that remove old bone matrix so that it can be replaced by new material are called**

A osteoblasts

B osteocytes

C osteoclasts✅

D macrophages

**206.Which of the following is NOT a hormone involved in the control of Ca2+ levels in plasma?**

A parathyroid hormone

B calcitonin

C ergocalciferol✅

D 1,25-dihydroxycholecalciferol

**207.In the regulation of Ca2+ balance**

A the parathyroid glands produce both parathyroid hormone and calcitonin

B ionized plasma Ca2+ controls parathyroid hormone release✅

C parathyroid hormone has a direct effect on bone causing bone reabsorption

D parathyroid hormone is a 64-amino-acid peptide and is the major controller of free calcium in the body

**208.Which of the following statements is NOT true?**
A Vitamin D is an umbrella term for two molecules, ergocalciferol and cholecalciferol
B The D vitamins are derived from dehydrocholesterol
C D vitamins are converted into 1,25-dihydroxycholecalciferol in the liver✅
D The major action of 1,25-dihydroxycholecalciferol is to enable Ca2+ absorption from the gut

**209.The following hormone(s) can promote the incorporation of calcium into bones**
A growth hormone
B thyroid hormone
C sex steroids
D all of the above✅

**210.Regarding the adrenal gland**
A The adrenal cortex produces steroids and catecholamines
B The zona glomerulosa produces aldosterone✅
C Adrenal androgen production is controlled by gonadotrophins
D Cortisol production is controlled by angiotensin

**211.Which of the following statements is NOT true?**
A The adrenal medulla is functionally part of the sympathetic nervous system
B Cushing's syndrome is due to overproduction of glucocorticoids
C The adrenal medulla and the adrenal cortex function independently from each other✅
D The chromaffin cells of the adrenal medulla manufacture and secrete adrenaline and noradrenaline

**212.Cortisol**
A is produced in the zona glomerulosa
B and its analogues are collectively classified as mineralocorticoids although they do have glucocorticoid actions
C release is stimulated by adrenocorticotropic hormone (ACTH)✅
D is not released during normal physiological activity but in response to stress

**213.Stress**
A response is driven by the amygdala which is part of the midbrain
B is the primary stimulus in the increased release of mineralocorticoids
C can cause increased activity in the thalamus
D can cause increased activity in the parasympathetic nerves that leads to acid secretion in the stomach✅

**214.Regarding the control of the reproductive system**
A gonadotrophin-releasing hormone is secreted by the pituitary
B the gonads secrete glucocorticoids
C testosterone is a steroid✅

D progesterone is secreted by the ovarian follicle

**215.Which of the following statements is true?**

A In the male, luteinizing hormone acts on the Sertoli cells to stimulate the production of testosterone

B Plasma luteinizing hormone levels are relatively constant minute to minute

C The hypothalamus secretes luteinizing hormone

D In the female, luteinizing hormone acts on the theca interna cells to stimulate the production of testosterone✅

**216.Leydig cells produce**

A mucus

B androgen-binding proteins

C testosterone✅

D FSH and LH

**217.During ovulation all of the following occur except**

A formation of the corpus luteum

B oestrogen production is very low✅

C high production of follicle-stimulating hormone

D high production of luteinizing hormone

**218.All of the following are involved in male sexual differentiation during early development, except**

A testosterone

B progesterone✅

C dihydrotestosterone

D Sry gene on the Y chromosome

**219.With regard to puberty**

A Puberty begins with the increased release of luteinizing hormone✅

B The onset of menstruation in females occurs at roughly 47 kg irrespective of age

C In males, luteinizing hormone stimulates the release of testosterone from the Sertoli cells

D Secondary sexual characteristics are stimulated by testosterone

**220.Which of the following statements is NOT true?**

A The penis becomes erect as a result of dilation of blood vessels entering the corpora cavernosum and corpus spongiosum

B In the penis, the parasympathetic nerves cause vasodilatation by releasing acetylcholine, vasoactive intestinal peptide and nitric oxide

C Nitric oxide increases the manufacture of cyclic guanosine monophosphate(cGMP) in blood vessel smooth muscle cells to cause them to contract✅

D Sildenafil (Viagra) inhibits the breakdown of cGMP and thus enhances erectile function

**221.An orgasm**

A is due to a reflex activation of the parasympathetic nerves in both males and females

B in males involves the mixing of the sperm and the secretions of the bulbourethral gland, seminal vesicle and prostate gland before they enter the urethra

C in males involves the secretion of the sugar glucose to provide energy

D in females may involve the release of pituitary oxytocin elicited by mechanical stimulation of the cervix✅

**222.The unfertilized ovum can survive for __ after ovulation**

A 1–2 h

B 10–12 h

C 24 h✅

D 2–3 days

**223.During fertilization**

A acid in the uterus triggers capacitation of sperm

B the glycoprotein ZP2 acts as a sperm receptor on the zona pellucida

C the ovum undergoes electrical depolarization and discharges granules that impair further binding of sperm✅

D after about 10 h the male and female pronuclei fuse

**224.Human chorionic gonadotrophin (hCG)**

A begins to be produced about 4 weeks after fertilization

B is produced by the corpus luteum after ovulation

C maintains the corpus luteum for a short period✅

D signals that a new menstrual cycle should begin

**225.Release of the hormone relaxin**

A inhibits the development of additional ova

B inhibits uterine contraction

C stimulates the development of the endometrium

D softens the cervix prior to parturition✅

**226.Regarding the control of lactation**

A prolactin stimulates milk ejection

B placental steroid secretion inhibits lactation during pregnancy✅

C vasopressin (ADH) is secreted in response to suckling

D oxytocin secretion can be suppressed by dopamine agonists

**227.Prolactin**

A inhibits release of follicle-stimulating hormone from the pituitary

B levels in the plasma rise following birth and lead to milk production

C is only released in and around birth

D release is maintained after birth mainly by suckling✅

**228.Colostrum**

A contains high levels of antibodies that provide the infant with basic immunological protection for the rest of its life

B is particularly rich in proteins✅

C has a higher sugar content than milk

D has a higher fat content than milk

**229.The main hormone involved in the milk let-down reflex is**

A progesterone

B oestrogens

C oxytocin✅

D prolactin

**230.Transduction describes**

A the transmission of signals across synapses by neurotransmitters

B the transmission of nerve impulses from receptors to cortex

C the translation of the stimulus into a code of action potentials✅

D the generation of conscious mental states from nerve impulses

**231.The modality of a sensory stimulus is encoded in the somatosensory system by**

A the type of receptor activated✅

B the frequency of action potentials evoked in the afferent nerve fibre

C the rate of receptor adaptation

D the method used to transduce the stimulus at the receptor

**232.A receptor potential is an example of**

A a frequency-coded signal

B a spatially coded signal

C a temporally coded signal

D an amplitude-coded signal✅

**233.Which of the following statements is NOT true?**

A Lateral inhibition is normally encountered under pathological conditions (e.g. strychnine poisoning)✅

B Inhibition can prevent the spread of excitation at progressively higher levels of the CNS by a phenomenon called lateral inhibition

C Lateral inhibition involves the excitation of inhibitory interneurones

D In almost all sensory systems, higher centres can exert inhibitory effects on all lower levels of the CNS

**234.Which of the following is NOT a sensory receptor found in mammals?**

A thermoreceptor

B pain receptor✅

C mechanoreceptor
D chemoreceptor

**235.A Ruffini-type receptor**
A is a rapidly adapting mechanoreceptor found in the non-hairy skin which responds to stretching of the skin
B is a slowly adapting mechanoreceptor found in non-hairy skin which is an intensity detector
C is a slowly adapting mechanoreceptor found in hairy skin which responds to stretching of the skin✅
D is a moderately rapidly adapting mechanoreceptor found in the hairy skin which responds to touch and pressure

**236.Thermoreceptors**
A respond to both warmth and cold
B respond to either warmth or cold✅
C are specialized nerve endings with corpuscular structures
D have a large receptive field (over 5 mm2)

**237.The conscious perception of pain**
A is associated with the activation of modality specific receptors
B may be delayed, despite the seriousness of the trauma
C may elicit autonomic reflexes of nausea, heavy sweating and lowered blood pressure
D all of the above✅

**238.Which tongue papillae are taste buds NOT associated with?**
A fungiform
B folliate
C circumvallate
D filiform✅

**239.The gustatory receptor cells in taste buds**
A have a lifespan of about 28 days
B are neuroepithelial cells arranged in a compact pear-shaped structure✅
C are all innervated by the same nerve fibre which branches to innervate all the cells in the taste bud
D are innervated by either the glossopharyngeal (IX) nerve or the trigeminal (V) nerve

**240.The two tissue layers that make up the olfactory organ are the**
A supporting cells and basal cells
B olfactory epithelium and lamina propria
C olfactory glands and olfactory receptors

D olfactory epithelium and olfactory receptors✅

**241.Humans can distinguish**
A over 10 000 different odours✅
B only about 100 different odours
C about 1000 different odours
D up to 5000 different odours

**242.In comparison to the cones, the rods are more**
A concentrated in the fovea region
B sensitive to dim light✅
C important for colour vision
D sensitive to detail

**243.Which of the following cells transmit impulses to the rest of the central nervous system via axons in the optic nerve?**
A ganglion cells✅
B bipolar cells
C amacrine cells
D horizontal cells

**244.Ganglion cells**
A are normally silent unless stimulated by light
B respond to a stationary light stimulus rather than to changes of light intensity
C can be divided into 50% ON-centre cells and 50% OFF-centre cells✅
D are further subdivided into P and M cells, where P cells are sensitive to contrast and movement and M cells are colour selective

**245.The axons of the ganglion cells from the __ region of the retina of the left eye and the __ region of the retina of the right eye proceed into the left optic tract**
A temporal, temporal
B nasal, nasal
C nasal, temporal
D temporal, nasal✅

**246.young, healthy human can detect sound wave frequencies of between**
A 10 Hz and 100 kHz
B 40 Hz and 50 kHz
C 40 Hz and 20 kHz✅
D 5 Hz and 20 kHz

**247.The order in which sound travels through the auditory system is**
A external auditory meatus, tympanic membrane, round window, scala tympani, scala vestibuli, oval window

B external auditory meatus, tympanic membrane, ossicles, oval window, scala tympani, scala media, round window
C external auditory meatus, tympanic membrane, ossicles, round window, scala vestibuli, scala tympani, oval window
D external auditory meatus, tympanic membrane, ossicles, oval window, scala vestibuli, scala tympani, round window✅

**248.The vestibular system is NOT involved in**
A balance
B hearing✅
C postural reflexes
D eye movements

**249.The utricle send signals representing __ movements and the saccule conveys information about __ movements**
A vertical, forward and backward
B forward and backward, vertical✅
C rotational, horizontal
D horizontal, rotational

**250.The idea of a voluntary movement is thought to originate in the**
A primary sensory cortex
B primary motor cortex
C association cortex✅
D cerebellum

**251.The motor cortex activates the α- and fusimotor neurones via the**
A lateral corticospinal tract
B corticorubrospinal tract
C neither a nor b
D both a and b✅

**252.Which of the following is NOT a functional and anatomical structure of the cerebellum?**
A spinocerebellum
B cerebrospinocerebellum✅
C cerebrocerebellum
D vestibulocerebellum

**253.The function of the cerebellum is**
A as a comparator, comparing sensory and motor inputs
B as a timing device, converting motor signals into a sequence of events
C as a store of motor information
D all of the above✅

**254.Which of the following mechanoreceptors is NOT a proprioceptor?**

A Golgi tendon organ
B muscle spindle
C Ruffini ending in joint
D Pacinian corpuscle✅

**255.Immediately following the section of both the dorsal and ventral nerve roots supplying a muscle, an increase in the frequency of discharge of a muscle spindle primary afferent from that muscle will be produced by**

A stimulation of the fusimotor fibres to the muscle distal to the section✅
B stimulation of the α motor neurone to the muscle distal to the section
C contraction of the homonymous muscle
D strong stimulation of the proximal end of the cut dorsal root

**256.The properties of muscle spindles and Golgi tendon organs make them most likely to be detectors of**

A position sense
B sense of movement
C force sensation✅
D all of the above

**257.What kind of reflexes are Golgi tendon organs involved in?**

A monosynaptic stretch reflexes
B withdrawal reflexes
C polysynaptic stretch reflexes
D protective reflex✅

***BEST OF LUCK***

***Mr. MOHAMED YAHIA ABUELGASEM KHATEM***